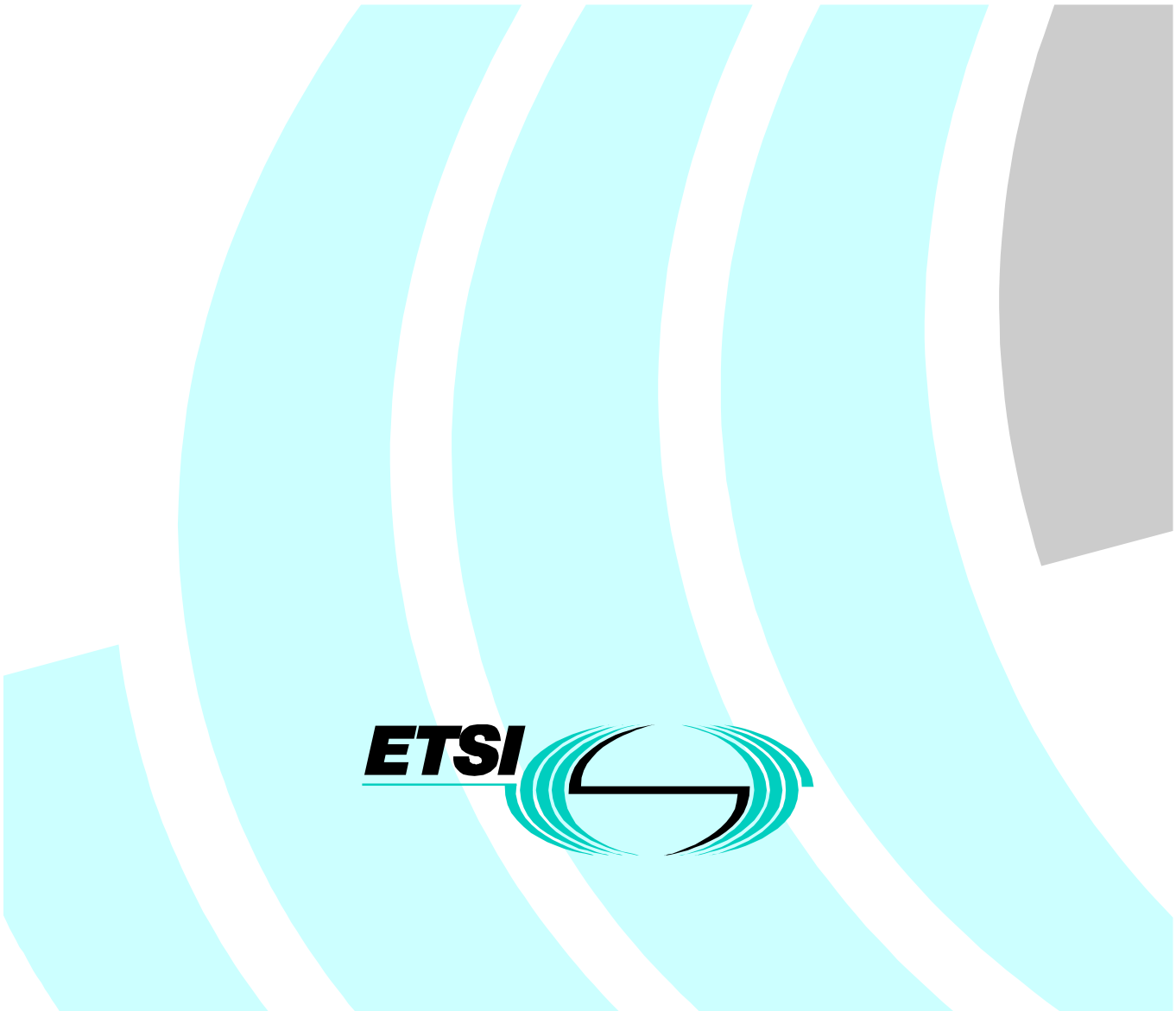
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Contents

Intellectual Property Rights 7

Foreword 7

1 Scope 8

2 References 8

3 Definitions, symbols and abbreviations 9
3.1 Definitions 9
3.2 Symbols 9
3.3 Abbreviations 9

4 General and operational requirements 9
4.1 Construction 9
4.2 Controls and indicators 10
4.3 Microphone and loudspeaker 11
4.4 Safety precautions 11
4.5 Labelling 11

5 Technical requirements 11
5.1 Switching time 11
5.2 Class of emission and modulation characteristics 11

6 General conditions of measurement 12
6.1 Arrangements for RF connections to the equipment 12
6.1.1 RF connections to integral antenna equipment 12
6.1.2 RF connection to equipment with a detachable antenna 12
6.2 Arrangements for test signals applied to the receiver input 12
6.3 Squelch 12
6.4 Normal test modulation 12
6.5 Artificial antenna 12
6.6 Arrangements for test signals applied to the transmitter input 12
6.7 Test channels 12
6.8 Measurement uncertainty and interpretation of the measured results 13
6.8.1 Measurement uncertainty 13
6.8.2 Interpretation of the measurement results 13
6.9 Test conditions, power sources and ambient temperatures 13
6.9.1 Normal and extreme test conditions 13
6.9.2 Test power source 14
6.10 Normal test conditions 14
6.10.1 Normal temperature and humidity 14
6.10.2 Normal power sources 14
6.10.2.1 Battery power source 14
6.10.2.2 Other power sources 14
6.11 Extreme test conditions 14
6.11.1 Extreme temperatures 14
6.11.2 Extreme values of test power sources 15
6.11.2.1 Battery power source 15
6.11.2.2 Other power sources 15
6.12 Procedure for tests at extreme temperatures 15

7 Environmental tests 15
7.1 Procedure 15
7.2 Performance check 15
7.3 Drop test 15
7.3.1 Definition 15
7.3.2 Method of measurement 16
7.3.3 Requirement 16
7.4 Temperature tests 16

7.4.1 Definition 16
7.4.2 Dry heat 16
7.4.2.1 Definition 16
7.4.2.2 Method of measurement 16
7.4.2.3 Requirement 16
7.4.3 Damp heat 16
7.4.3.1 Definition 16
7.4.3.2 Method of measurement 17
7.4.3.3 Requirement 17
7.4.4 Low temperature cycle 17
7.4.4.1 Definition 17
7.4.4.2 Method of measurement 17
7.4.4.3 Requirement 17

8 Transmitter 17
8.1 Frequency error 17
8.1.1 Definition 17
8.1.2 Method of measurement 18
8.1.3 Limits 18
8.2 Carrier power 18
8.2.1 Definitions 18
8.2.2 Method of measurement 18
8.2.3 Limits, Normal and extreme test conditions 18
8.3 Frequency deviation 18
8.3.1 Definition 18
8.3.2 Maximum permissible frequency deviation 18
8.3.2.1 Method of measurement 18
8.3.2.2 Limits 19
8.3.3 Reduction of frequency deviation at modulation frequencies above 3 kHz 19
8.3.3.1 Method of measurement 19
8.3.3.2 Limits 19
8.4 Sensitivity of the modulator, including microphone 19
8.4.1 Definition 19
8.4.2 Method of measurement 20
8.4.3 Limits 20
8.5 Audio frequency response 20
8.5.1 Definition 20
8.5.2 Method of measurement 20
8.5.3 Limit 20
8.6 Audio frequency harmonic distortion of the emission 21
8.6.1 Definition 21
8.6.2 Method of measurement 21
8.6.2.1 Normal test conditions 21
8.6.2.2 Extreme test conditions 22
8.6.3 Limits 22
8.7 Adjacent channel power 22
8.7.1 Definition 22
8.7.2 Method of measurement 22
8.7.3 Limits 22
8.8 Conducted spurious emissions conveyed to the antenna 23
8.8.1 Definition 23
8.8.2 Method of measurement 23
8.8.3 Limit 23
8.9 Cabinet radiation and conducted spurious emissions other than those conveyed to the antenna 23
8.9.1 Definitions 23
8.9.2 Method of measurement 23
8.9.3 Limits 24
8.10 Residual modulation of the transmitter 24
8.10.1 Definition 24
8.10.2 Method of measurement 25
8.10.3 Limit 25

8.11 Transient frequency behaviour of the transmitter 25
8.11.1 Definitions 25
8.11.2 Method of measurement 26
8.11.3 Limits 27

9 Receiver 29
9.1 Harmonic distortion and rated audio frequency output power 29
9.1.1 Definition 29
9.1.2 Methods of measurement 29
9.1.3 Limits 29
9.2 Audio frequency response 29
9.2.1 Definition 29
9.2.2 Method of measurement 29
9.2.3 Limits 30
9.3 Maximum usable sensitivity 31
9.3.1 Definition 31
9.3.2 Method of measurement 31
9.3.3 Limits 31
9.4 Co-channel rejection 31
9.4.1 Definition 31
9.4.2 Method of measurement 31
9.4.3 Limit 32
9.5 Adjacent channel selectivity 32
9.5.1 Definition 32
9.5.2 Method of measurement 32
9.5.3 Limits 32
9.6 Spurious response rejection 32
9.6.1 Definition 32
9.6.2 Method of measurement 32
9.6.3 Limit 33
9.7 Intermodulation response 33
9.7.1 Definition 33
9.7.2 Method of measurement 33
9.7.3 Limit 33
9.8 Blocking or desensitization 33
9.8.1 Definition 33
9.8.2 Method of measurement 34
9.8.3 Limit 34
9.9 Conducted spurious emissions 34
9.9.1 Definition 34
9.9.2 Method of measurement 34
9.9.3 Limit 34
9.10 Radiated spurious emissions 34
9.10.1 Definition 34
9.10.2 Method of measurements 35
9.10.3 Limit 35
9.11 Receiver noise and hum level 36
9.11.1 Definition 36
9.11.2 Method of measurement 36
9.11.3 Limit 36
9.12 Squelch operation 36
9.12.1 Definition 36
9.12.2 Method of measurement 36
9.12.3 Limits 37
9.13 Squelch hysteresis 37
9.13.1 Definition 37
9.13.2 Method of measurement 37
9.13.3 Limit 37

Annex A (normative): Measuring receiver for adjacent channel power measurement 38

A.1 Power measuring receiver specification 38
A.1.1 IF filter 38
A.1.2 Attenuation indicator 39
A.1.3 Rms value indicator 39
A.1.4 Oscillator and amplifier 39

Annex B (normative): Radiated measurements 40

B.1 Test sites and general arrangements for measurements involving the use of radiated fields 40
B.1.1 Outdoor test site 40
B.1.2 Test antenna 40
B.1.3 Substitution antenna 41
B.1.4 Optional additional indoor site 41

B.2 Guidance on the use of radiation test sites 42
B.2.1 Measuring distance 42
B.2.2 Test antenna 42
B.2.3 Substitution antenna 42
B.2.4 Artificial antenna 43
B.2.5 Auxiliary cables 43
B.2.6 Acoustic measuring arrangement 43

B.3 Further optional alternative indoor test site using an anechoic chamber 43
B.3.1 Example of the construction of a shielded anechoic chamber 44
B.3.2 Influence of parasitic reflections in anechoic chambers 44
B.3.3 Calibration of the shielded anechoic chamber 45

History 47

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Foreword

This European Standard (Telecommunications series) has been produced by ETSI Technical Committee Electromagnetic compatibility and Radio spectrum Matters (ERM).

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1 Scope

The present document states the minimum technical characteristics and methods of measurement required for portable Very High Frequency (VHF) radiotelephones not providing maritime distress and safety communications functions (i.e. not forming part of the Global Maritime Distress and Safety System (GMDSS)) operating in the bands between 156 MHz and 174 MHz allocated to the maritime mobile services by the Radio Regulations, Appendix 18 [1].

The present document also specifies technical characteristics, methods of measurement and required test results.

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies.
- A non-specific reference to an ETS shall also be taken to refer to later versions published as an EN with the same number.

[1] Radio Regulations, Appendix 18 (1990): "Table of Transmitting Frequencies in the Band 156-174 MHz for Stations in the Maritime Mobile Service".

[2] ITU-T Recommendation E.161 (1993): "Arrangement of digits, letters and symbols on telephones and other devices that can be used for gaining access to a telephone network".

[3] ITU-R Recommendation M.493-9 (1997): "Digital selective-calling system for use in the maritime mobile service".

[4] ITU-R Recommendation M.541-8 (1997): "Operational procedures for the use of digital selective-calling equipment in the maritime mobile service".

[5] ETS 300 225: "Radio Equipment and Systems (RES); Technical characteristics and methods of measurement for survival craft portable VHF radiotelephone apparatus".

[6] ITU-T Recommendation P.53 (1994): "Psophometer for use on telephone-type circuits".

[7] ITU-R Recommendation SM 332-4: "Selectivity of receivers".

[8] ETR 028: "Radio Equipment and Systems (RES); Uncertainties in the measurement of mobile radio equipment characteristics".

3 Definitions, symbols and abbreviations

3.1 Definitions

For the purposes of the present document, the following definitions apply:

integral antenna: antenna that is permanently fixed to the equipment and not detachable by the user.

detachable antenna: antenna fixed to the equipment by means of an antenna connector and detachable by the user.

G3E: phase-modulation (frequency modulation with a pre-emphasis of 6 dB/octave) for speech.

modulation index: ratio between the frequency deviation and the modulation frequency.

performance check: check of:

- the transmitter carrier power and frequency; and
- receiver sensitivity.

3.2 Symbols

For the purposes of the present document, the following symbols apply:

dBA	Relative to 2×10^{-5} Pa
emf	Electromotive force

3.3 Abbreviations

For the purposes of the present document, the following abbreviations apply:

DSC	Digital Selective Calling
EUT	Equipment Under Test
GMDSS	Global Maritime Distress and Safety System
rms	root mean square
SINAD	(Signal + Noise + Distortion)/(Noise + Distortion)
VHF	Very High Frequency

4 General and operational requirements

4.1 Construction

The manufacturer shall declare that compliance to the requirements of clause 4 is achieved and shall provide relevant documentation.

The mechanical and electrical construction and finish of the equipment shall conform in all respects to good engineering practice, and the equipment shall be suitable for use on board ships.

All controls shall be of sufficient size to enable the usual control functions to be easily performed and the number of controls should be the minimum necessary for simple and satisfactory operation.

All parts of the equipment to be checked during inspection or maintenance operations shall be readily accessible. The components shall be readily identifiable.

Technical documentation shall be supplied with the equipment.

The VHF maritime mobile service uses both single-frequency and two-frequency channels. For two-frequency channels there shall be a separation of 4,6 MHz between the transmitting frequency and the receiving frequency (see Radio Regulations Appendix 18 [1]).

The equipment, which can consist of more than one unit, shall be capable of operating on single frequency and two-frequency channels with manual control (simplex).

The equipment shall be of a colour which distinguishes it from the portable VHF equipment specified in ETS 300 225 [5].

The equipment shall be able to operate on all channels defined in Radio Regulations, Appendix 18 [1].

Operation on channels 75 and 76 shall be restricted to a maximum transmitted power of 1W.

Additional VHF channels outside those defined by Radio Regulations, Appendix 18 [1] may also be

provided, but means shall be provided to block any channel, including Appendix 18 channels.

It shall not be possible for the user to unblock or block any channels.

Scan or multiple watch may be provided but means shall be provided to block or unblock these functions.

If the equipment is fitted with an auxiliary antenna connector, simultaneous connection of both the auxiliary antenna and the normal antenna shall be prevented.

The equipment shall be so designed that use of channel 70 for purposes other than Digital Selective Calling (DSC) is prevented (see ITU-R Recommendation M.493-9 [3] and ITU-R Recommendation M.541–8 [4]).

It shall not be possible to transmit while any frequency synthesizer used within the transmitter is out of lock.

It shall not be possible to transmit during channel switching operations.

4.2 Controls and indicators

The equipment shall have a channel selector and shall indicate the designator, as shown in Radio Regulations, Appendix 18 [1], of the channel at which the equipment is set. The channel designator shall be legible irrespective of the external lighting conditions.

Where an input panel on the equipment for entering the digits 0 to 9 is provided, this shall conform to ITU-T Recommendation E.161 [2].

The equipment shall have the following additional controls and indicators:

- on/off switch for the equipment with a visual indication that the equipment is in operation;
- a manual, non-locking push to talk switch to operate the transmitter;
- a switch for reducing transmitter output power to no more than 1 W where the RF output power is more than 1 watt;
- an audio frequency power volume control;
- a squelch control;
- a visual indication that the transmitter is activated.

The equipment shall also meet the following requirements:

- the user shall not have access to any control which, if wrongly set, might impair the technical characteristics of the equipment.

4.3 Microphone and loudspeaker

The equipment shall be fitted with an integral microphone and an integral loudspeaker.

During transmission the receiver output shall be muted.

4.4 Safety precautions

Measures shall be taken to protect the equipment against the effects of overcurrent or overvoltage.

Measures shall be taken to prevent damage to the equipment that might arise from an accidental reversal of polarity of the electrical power source.

No damage to the equipment shall occur when the antenna terminals are placed on open circuit or short circuit while transmitting for a period of at least 5 minutes in each case.

In order to provide protection against damage due to the build up of static voltages at the antenna terminals, there shall be a dc path from the antenna terminals to chassis not exceeding 100 kΩ.

The information in any volatile memory device shall be protected from interruptions in the power supply of up to 60 s duration.

4.5 Labelling

All controls, instruments, indicators and terminals shall be clearly labelled.

Details of any external power supply from which the equipment is intended to operate shall be clearly indicated on the equipment.

The equipment shall be clearly and indelibly marked on the exterior with the identification of the manufacturer, type designation of the equipment, the serial number of the unit and the text: "Not intended for distress and safety purposes".

The compass safe distance shall be stated on the equipment.

5 Technical requirements

5.1 Switching time

The channel switching arrangement shall be such that the time necessary to change over from using one of the channels to using any other channel does not exceed 5 s.

The time necessary to change over from transmission to reception or vice versa, shall not exceed 0,3 s.

5.2 Class of emission and modulation characteristics

The equipment shall use phase modulation, G3E (frequency modulation with a pre-emphasis of 6 dB/octave) for speech.

The equipment shall be designed to operate satisfactorily with a channel separation of 25 kHz.

The frequency deviation corresponding to 100 % modulation shall be ± 5 kHz as nearly as practicable.

6 General conditions of measurement

6.1 Arrangements for RF connections to the equipment

6.1.1 RF connections to integral antenna equipment

For equipment without an antenna connector, the manufacturer shall prepare the equipment with a temporary 50 ohm connector to be used as the RF input/output port.

6.1.2 RF connection to equipment with a detachable antenna

Equipment having an antenna connector shall be tested using the antenna connector as the RF input/output port.

In the case where equipment has more than one antenna connector, the connector normally used to connect the portable antenna to the equipment shall be used.

6.2 Arrangements for test signals applied to the receiver input

Test signal sources shall be connected to the receiver input in such a way that the impedance presented to the receiver input is 50 Ω, irrespective of whether one or more test signals are applied to the receiver simultaneously.

The levels of the test signals shall be expressed in terms of the emf at the terminals to be connected to the receiver.

The nominal frequency of the receiver is the carrier frequency of the selected channel.

6.3 Squelch

Unless otherwise specified, the receiver squelch facility shall be made inoperative for the duration of the conformance tests.

6.4 Normal test modulation

For normal test modulation, the modulation frequency shall be 1 kHz and the frequency deviation shall be ± 3 kHz.

6.5 Artificial antenna

When tests are carried out with an artificial antenna, this shall be a non-reactive, non-radiating 50 Ω load.

6.6 Arrangements for test signals applied to the transmitter input

For the purpose of the present document, the audio frequency modulating signal applied to the transmitter shall be produced by a signal generator applied to the connection terminals replacing the microphone transducer.

6.7 Test channels

Tests shall be made on channel 16 unless otherwise stated.

6.8 Measurement uncertainty and interpretation of the measured results

6.8.1 Measurement uncertainty

Table 1: Absolute measurement uncertainties: maximum values

Parameter	Maximum uncertainty
RF frequency	± 1 x 10^{-7}
RF power	± 0,75 dB
Maximum frequency deviation: - within 300 Hz to 6 kHz of audio frequency - within 6 kHz to 25 kHz of audio frequency	 ± 5 % ± 3 dB
Deviation limitation	± 5 %
Adjacent channel power	± 5 dB
Conducted spurious emission of transmitter	± 4 dB
Audio output power	± 0,5 dB
Amplitude characteristics of receiver limiter	± 1,5 dB
Sensitivity at 20 dB SINAD	± 3 dB
Conducted emission of receiver	± 3 dB
Two-signal measurement	± 4 dB
Three-signal measurement	± 3 dB
Radiated emission of transmitter	± 6 dB
Radiated emission of receiver	± 6 dB
Transmitter transient time	± 20 %
Transmitter transient frequency	± 250 Hz
Receiver desensitization (duplex operation)	± 0,5 dB

For the test methods according to the present document the uncertainty figures are valid to a confidence level of 95 % calculated according to the methods described in ETR 028 [8].

6.8.2 Interpretation of the measurement results

The interpretation of the results recorded in a test report for the measurements described in the present document shall be as follows:

- the measured value related to the corresponding limit will be used to decide whether an equipment meets the requirements of the present document;
- the measurement uncertainty value for the measurement of each parameter shall be included in the test report;
- the recorded value of the measurement uncertainty shall be, for each measurement, equal to or lower than the figures in table 1 above.

NOTE: This procedure for using maximum acceptable uncertainty values is valid until superseded by other appropriate publications of ETSI covering this subject.

6.9 Test conditions, power sources and ambient temperatures

6.9.1 Normal and extreme test conditions

Conformance tests shall be performed under normal test conditions and also, where stated, under extreme test conditions (subclause 6.11.1 and subclause 6.11.2 applied simultaneously).

6.9.2 Test power source

During conformance testing, the equipment shall be supplied from a test power source capable of producing normal and extreme test voltages as specified in subclause 6.10.2 and subclause 6.11.2.

The internal impedance of the test power source shall be low enough for its effect on the test results to be negligible. For the purpose of testing the power source voltage shall be measured at the input terminals of the equipment.

During testing, the power source voltages shall be maintained within a tolerance of ± 3 % relative to the voltage level at the beginning of each test.

6.10 Normal test conditions

6.10.1 Normal temperature and humidity

The normal temperature and humidity conditions for tests shall be a combination of temperature and humidity within the following ranges:

- temperature: +15°C to +35°C;
- relative humidity: 20 % to 75 %.

When the relative humidity is lower than 20 %, it shall be stated in the test report.

6.10.2 Normal power sources

6.10.2.1 Battery power source

Where the equipment is designed to operate from a battery, the normal test voltage shall be the nominal voltage of the battery as declared by the manufacturer.

6.10.2.2 Other power sources

For operation from other power sources the normal test voltage shall be that declared by the manufacturer.

6.11 Extreme test conditions

Unless otherwise stated the extreme tests conditions means that the Equipment Under Test (EUT) shall be tested at the upper temperature and at the upper limit of the supply voltage applied simultaneously, and at the low temperature and the lower limit of the supply voltage applied simultaneously.

6.11.1 Extreme temperatures

For tests at extreme temperatures, measurements shall be made in accordance with subclause 6.12, at a lower temperature of -15°C and an upper temperature of +55°C.

6.11.2 Extreme values of test power sources

6.11.2.1 Battery power source

The upper extreme test voltage shall be the terminal voltage of the battery (fresh primary battery or fully charged secondary battery) when loaded by the equipment at normal temperature in the receive condition with the squelch operated to mute the audio.

The lower extreme test voltage shall be 0,85 times the value determined above.

Where equipment can be powered by batteries of differing terminal voltage then the upper extreme test voltage shall be determined using the highest terminal voltage battery and the lower extreme test voltage shall be 0,85 times the upper extreme of the lowest terminal voltage battery.

The test report shall note details of all batteries so assessed.

6.11.2.2 Other power sources

For operation from other power sources the extreme test voltages shall be agreed between the testing authority and the equipment manufacturer.

6.12 Procedure for tests at extreme temperatures

The equipment shall be switched off during the temperature stabilizing periods.

Before conducting tests at the upper temperature, the equipment shall be placed in the test chamber and left until thermal equilibrium is reached. The equipment shall then be switched on for half an hour in the high power transmit condition at the normal voltage, the equipment shall meet the requirement of the present document.

For tests at the lower temperature, the equipment shall be left in the test chamber until thermal equilibrium is reached and shall then be switched to the standby or receive position for one minute, after which the equipment shall meet the requirements of the present document.

7 Environmental tests

7.1 Procedure

Environmental tests shall be carried out before testing the same equipment to the other requirements of the present document. Unless otherwise stated, the equipment shall be connected to an electrical power source during the periods for which it is specified that electrical tests shall be carried out. These tests shall be performed using the normal test voltage.

7.2 Performance check

A performance check shall be a check of transmitter frequency error, subclause 8.1, transmitter carrier power, subclause 8.2 and maximum usable sensitivity, subclause 9.3. These performance checks shall only be performed under normal test conditions.

7.3 Drop test

7.3.1 Definition

This test simulates the effects of a free fall of the equipment onto the deck of a ship resulting from mishandling.

7.3.2 Method of measurement

The test shall consist of a series of 6 drops, one on each surface of the equipment.

The test shall be carried out under normal temperature and humidity.

The test surface shall consist of a piece of solid hard wood with a thickness of at least 150 mm and a mass of 30 kg or more.

The height of the lowest part of the equipment relative to the test surface at the moment of release shall be 1 000 mm ± 10 mm.

The equipment shall be subjected to this test configured for use as in operational circumstances.

At the end of the test the equipment shall be subjected to a performance check and shall then be examined for external indications of damage. The findings shall be noted in the test report.

7.3.3 Requirement

The equipment shall meet the requirements of the performance check.

There shall be no harmful deterioration of the equipment visible.

7.4 Temperature tests

7.4.1 Definition

The immunity against the effects of temperature is the ability of the equipment to maintain the specified mechanical and electrical performance after the following tests has been carried out.

7.4.2 Dry heat

7.4.2.1 Definition

This test determines the ability of equipment to be operated at high ambient temperatures and to operate through temperature changes.

7.4.2.2 Method of measurement

The EUT shall be placed in a chamber at normal room temperature and relative humidity. The EUT and, if appropriate, any climatic control devices with which it is provided shall then be switched on. The temperature shall then be raised to and maintained at +55°C ± 3°C.

At the end of a soak period of 10 to 16 hours at +55°C ± 3°C, the EUT shall be subjected to the performance check. The temperature of the chamber shall be maintained at +55°C ± 3°C during the whole performance check period. At the end of the test, the EUT shall be returned to normal environmental conditions or to those required at the start of the next test. The maximum rate of raising or reducing the temperature of the chamber in which the equipment is being tested shall be 1°C/minute.

7.4.2.3 Requirement

The equipment shall meet the requirements of the performance check.

7.4.3 Damp heat

7.4.3.1 Definition

This test determines the ability of equipment to be operated under conditions of high humidity.

7.4.3.2 Method of measurement

The EUT shall be placed in a chamber at normal room temperature and relative humidity. The temperature shall then be raised to +40°C ± 2°C, and the relative humidity raised to 93 % ± 3 % over a period of 3 hours ± 0,5 hour. These conditions shall be maintained for a period of 10 to 16 hours. Any climatic control devices provided in the EUT may be switched on at the conclusion of this period. The EUT shall be switched on 30 minutes later, or after such period as agreed by the manufacturer, and shall be kept operational for at least 2 hours during which period the EUT shall be subjected to the performance check once. The temperature and relative humidity of the chamber shall be maintained as specified during the whole test period. At the end of the test period and with the EUT still in the chamber, the chamber shall be brought to room temperature in not less than 1 hour. At the end of the test the EUT shall be returned to normal environmental conditions or to those required at the start of the next test. The maximum rate of raising or reducing the temperature of the chamber in which the equipment is being tested shall be 1°C/minute.

7.4.3.3 Requirement

The equipment shall meet the requirements of the performance check.

7.4.4 Low temperature cycle

7.4.4.1 Definition

These tests determine the ability of equipment to be operated at low temperatures. They also allow equipment to demonstrate an ability to start up at low ambient temperatures.

7.4.4.2 Method of measurement

The EUT shall be placed in a chamber at normal room temperature and relative humidity. The temperature shall then be reduced to and maintained at -15°C ± 3°C, for a period of 10 to 16 hours. Any climatic control devices provided in the EUT may be switched on at the conclusion of this period. The EUT shall be switched on 30 minutes later, and shall be kept operational for at least 2 hours during which period the EUT shall be subjected to the performance check once. The temperature of the chamber shall be maintained at -15°C ± 3°C during the whole test period. At the end of the test the EUT shall be returned to normal environmental conditions or to those required at the start of the next test. The maximum rate of raising or reducing the temperature of the chamber in which the equipment is being tested shall be 1°C/minute.

7.4.4.3 Requirement

The equipment shall meet the requirements of the performance check.

8 Transmitter

All tests on the transmitter shall be carried out with the output power switch set at its maximum except where otherwise stated.

8.1 Frequency error

8.1.1 Definition

The frequency error is the difference between the measured carrier frequency and its nominal value.

8.1.2 Method of measurement

The carrier frequency shall be measured in the absence of modulation, with the transmitter connected to an artificial antenna (subclause 6.5). Measurements shall be made under normal test conditions (subclause 6.10) and under extreme test conditions (subclause 6.11).

This test shall be carried out with the output power switch being set at both maximum and minimum.

8.1.3 Limits

The frequency error shall be within ± 1,5 kHz.

8.2 Carrier power

8.2.1 Definitions

The carrier power is the mean power delivered to the artificial antenna during one radio frequency cycle in the absence of modulation.

The rated output power is the carrier power declared by the manufacturer.

8.2.2 Method of measurement

The transmitter shall be connected to an artificial antenna (subclause 6.5) and the power delivered to this artificial antenna shall be measured. The measurements shall be made under normal test conditions (subclause 6.10) and also under extreme test conditions (subclause 6.11) on the highest frequency channel, the lowest frequency channel and on channel 16.

8.2.3 Limits, Normal and extreme test conditions

The carrier power on the Appendix 18 channels, Radio Regulations, Appendix 18 [1], with the output power switch (subclause 4.2) set at maximum, shall not exceed 6 W.

With the output power switch at minimum, the carrier power shall remain between 0,1 W and 1 W.

8.3 Frequency deviation

8.3.1 Definition

For the purpose of the present document, the frequency deviation is the difference between the instantaneous frequency of the modulated radio frequency signal and the carrier frequency.

8.3.2 Maximum permissible frequency deviation

8.3.2.1 Method of measurement

The frequency deviation shall be measured at the output with the transmitter connected to an artificial antenna (subclause 6.5), by means of a deviation meter capable of measuring the maximum deviation, including that due to any harmonics and intermodulation products which may be generated in the transmitter.

The modulation frequency shall be varied between 300 Hz and 3 kHz. The level of this test signal shall be 20 dB above the level which produces normal test modulation (subclause 6.4). This test shall be repeated with the output power switch set at maximum and minimum.

8.3.2.2 Limits

The maximum permissible frequency deviation shall be ± 5 kHz.

8.3.3 Reduction of frequency deviation at modulation frequencies above 3 kHz

8.3.3.1 Method of measurement

The transmitter shall be operated under normal test conditions (subclause 6.10) connected to a load as specified in subclause 6.5. The transmitter shall be modulated by the normal test modulation (subclause 6.4). With the input level of the modulation signal being kept constant, the modulation frequency shall be varied between 3 kHz and 25 kHz and the frequency deviation shall be measured.

8.3.3.2 Limits

For modulation frequencies between 3 kHz and 6 kHz the frequency deviation shall not exceed the frequency deviation with a modulation frequency of 3 kHz. For a modulation frequency of 6 kHz, the frequency deviation shall not exceed ± 1,5 kHz, as shown in figure 1.

For modulation frequencies between 6 kHz and 25 kHz, the frequency deviation shall not exceed that given by a linear response of frequency deviation (in dB) against modulation frequency, starting at the point where the modulation frequency is 6 kHz and the frequency deviation is ± 1,5 kHz and inclined at 14 dB per octave, with the frequency deviation diminishing as the modulation frequency increases, as shown in figure 1.

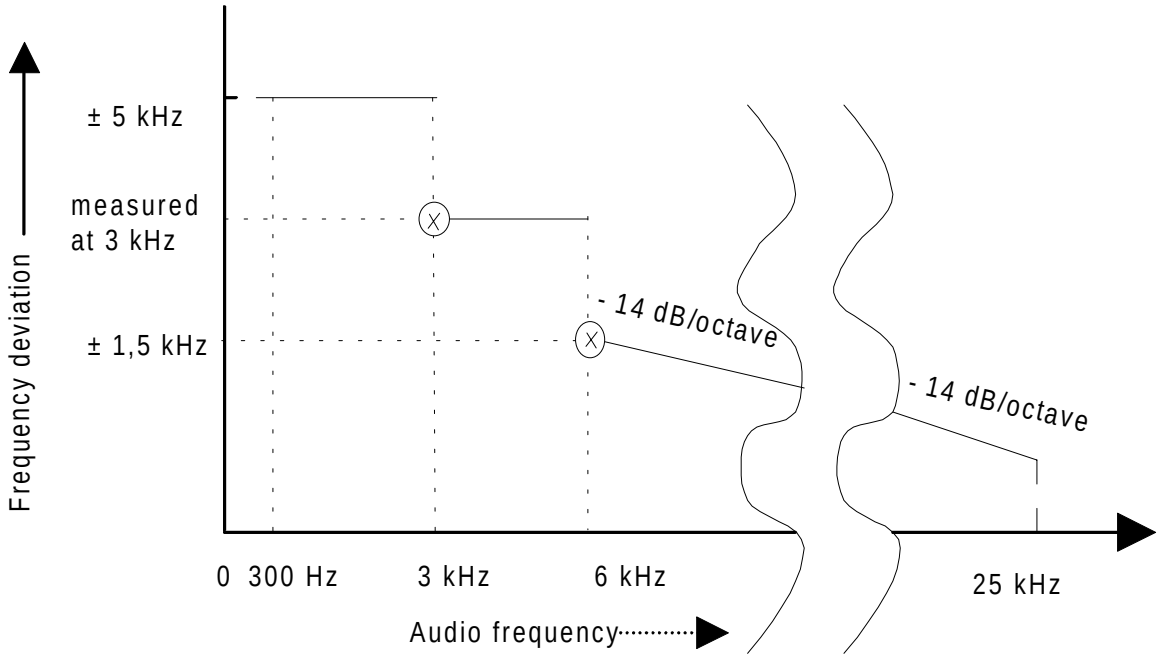


Figure 1

8.4 Sensitivity of the modulator, including microphone

8.4.1 Definition

This characteristic expresses the capability of the transmitter to produce sufficient modulation when an audio frequency signal corresponding to the normal mean speech level is applied to the microphone.

8.4.2 Method of measurement

An acoustic signal with a frequency of 1 kHz and sound level of 94 dBA shall be applied to the microphone. The resulting deviation shall be measured.

8.4.3 Limits

The resulting frequency deviation shall be between ± 1,5 kHz and ± 3 kHz.

8.5 Audio frequency response

8.5.1 Definition

The audio frequency response is the frequency deviation of the transmitter as a function of the modulating frequency.

8.5.2 Method of measurement

A modulating signal at a frequency of 1 kHz shall be applied to the transmitter and the deviation shall be measured at the output. The audio input level shall be adjusted so that the frequency deviation is ± 1 kHz. This is the reference point in figure 2 (1 kHz corresponds to 0 dB).

The modulation frequency shall then be varied between 300 Hz and 3 kHz, with the level of the audio frequency signal being kept constant and equal to the value specified above.

The test shall be carried out on one channel only (see subclause 6.7).

8.5.3 Limit

The audio frequency response shall be within +1 dB and -3 dB of a 6 dB/octave line passing through the reference point (see figure 2).

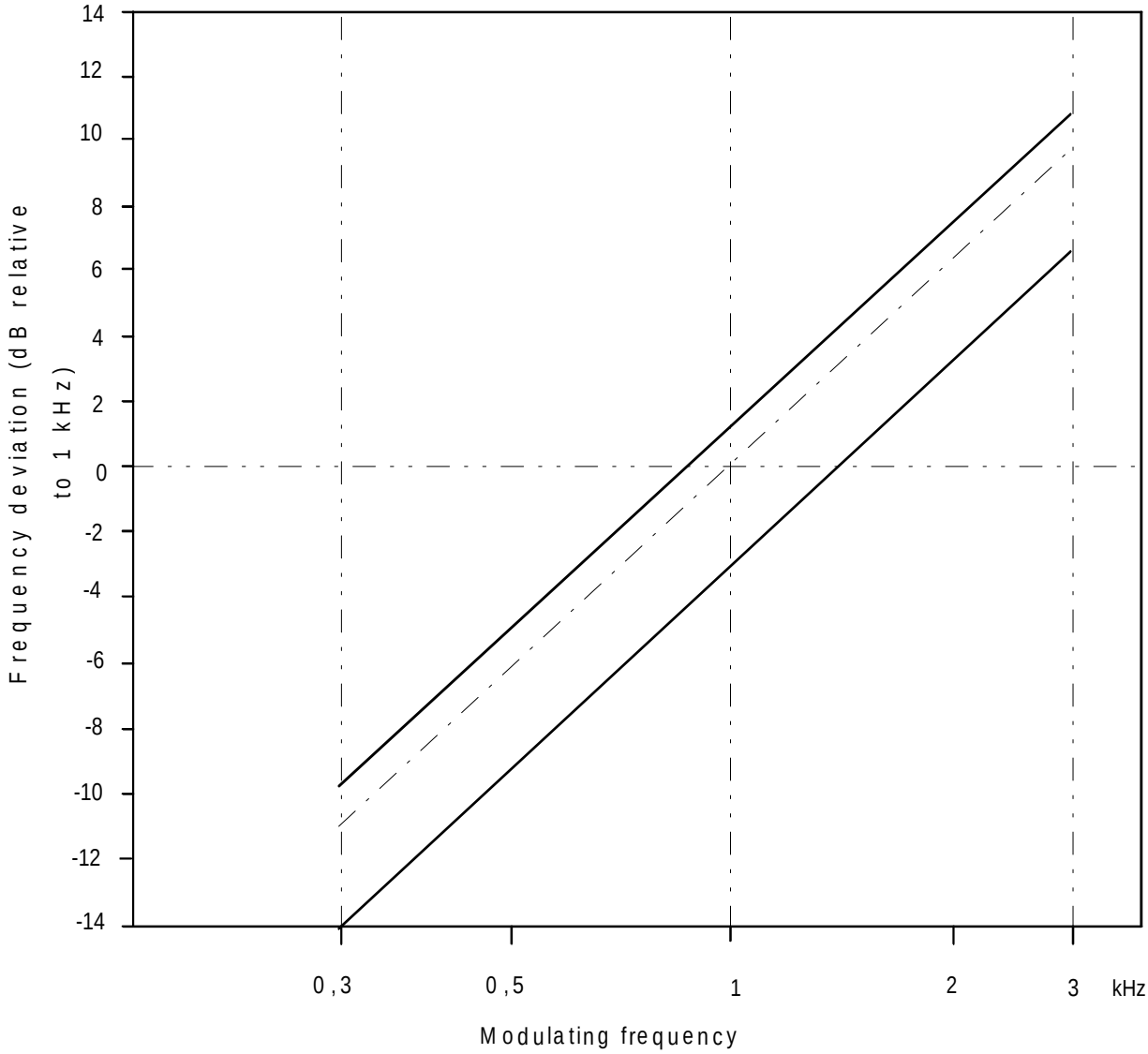


Figure 2: Audio frequency response

8.6 Audio frequency harmonic distortion of the emission

8.6.1 Definition

The harmonic distortion of the emission modulated by an audio frequency signal is defined as the ratio, expressed as a percentage, of the root mean square (rms) voltage of all the harmonic components of the fundamental modulation frequency to the total rms voltage of the modulation signal after linear demodulation.

8.6.2 Method of measurement

The RF signal produced by the transmitter shall be applied via an appropriate coupling device to a linear demodulator with a de-emphasis network of 6 dB per octave. This test shall be carried out with the output power switch at both maximum and minimum.

8.6.2.1 Normal test conditions

Under normal test conditions (subclause 6.10) the RF signal shall be modulated successively at frequencies of 300 Hz, 500 Hz and 1 kHz with a constant modulation index of 3.

The distortion of the audio frequency signal shall be measured at all the frequencies specified above.

8.6.2.2 Extreme test conditions

Under extreme test conditions (subclause 6.11.1 and subclause 6.11.2 applied simultaneously), the measurements shall be carried out at 1 kHz with a frequency deviation of ± 3 kHz.

8.6.3 Limits

The harmonic distortion shall not exceed 10 %.

8.7 Adjacent channel power

8.7.1 Definition

The adjacent channel power is that part of the total power output of a transmitter under defined conditions of modulation, which falls within a specified passband centred on the nominal frequency of either of the adjacent channels. This power is the sum of the mean power produced by the modulation, hum and noise of the transmitter.

8.7.2 Method of measurement

The test shall performed on the lowest frequency channel, the highest frequency channel and on channel 16.

The adjacent channel power can be measured with a power measuring receiver which conforms to annex A (referred to in subclause 8.7.2 and annex A as the "receiver") see also ITU-T Recommendation SM 332-4 [7]:

a) The transmitter shall be operated at the carrier power determined in subclause 8.2 under normal test conditions. The output of the transmitter shall be linked to the input of the "receiver" by a connecting device such that the impedance presented to the transmitter is 50 Ω and the level at the "receiver" input is appropriate.

b) With the transmitter unmodulated, the tuning of the "receiver" shall be adjusted so that a maximum response is obtained. This is the 0 dB response point. The "receiver" attenuator setting and the reading of the meter shall be recorded.

   The measurement may be made with the transmitter modulated with normal test modulation, in which case this fact shall be recorded with the test results.

c) The tuning of the "receiver" shall be adjusted away from the carrier so that the "receiver" -6 dB response nearest to the transmitter carrier frequency is located at a displacement from the nominal carrier frequency of 17 kHz.

d) The transmitter shall be modulated with 1,25 kHz at a level which is 20 dB higher than that required to produce ± 3 kHz deviation.

e) The "receiver" variable attenuator shall be adjusted to obtain the same meter reading as in step b) or a known relation to it.

f) The ratio of adjacent channel power to carrier power is the difference between the attenuator settings in steps b) and e), corrected for any differences in the reading of the meter.

g) The measurement shall be repeated with the "receiver" tuned to the other side of the carrier.

8.7.3 Limits

The adjacent channel power shall not exceed a value of 70 dB below the carrier power of the transmitter without any need to be below 0,2 µW.

8.8 Conducted spurious emissions conveyed to the antenna

8.8.1 Definition

Conducted spurious emissions are emissions on a frequency or frequencies which are outside the necessary bandwidth and the level of which may be reduced without affecting the corresponding transmission of information. Spurious emissions include harmonic emissions, parasitic emissions, intermodulation products and frequency conversion products, but exclude out of band emissions.

8.8.2 Method of measurement

Conducted spurious emissions shall be measured with the unmodulated transmitter connected to the artificial antenna (subclause 6.5).

The measurements shall be made over a range from 9 kHz to 2 GHz, excluding the channel on which the transmitter is operating and its adjacent channels.

The measurements for each spurious emission shall be made using a tuned radio measuring instrument or a spectrum analyser.

8.8.3 Limit

The power of any conducted spurious emission on any discrete frequency shall not exceed 0,25 µW.

8.9 Cabinet radiation and conducted spurious emissions other than those conveyed to the antenna

8.9.1 Definitions

Cabinet radiation consists of emissions at frequencies, other than those of the carrier and the sideband components resulting from the wanted modulation process, which are radiated by the equipment cabinet and structures.

Conducted spurious emissions other than those conveyed to the antenna are emissions at frequencies, other than those of the carrier and the sideband components resulting from the wanted modulation process, which are produced by conduction in the wiring and accessories used with the equipment.

Integral antenna equipment shall be tested with the normal antenna fitted and the carrier frequency emission shall be filtered as described in the method of measurement.

8.9.2 Method of measurement

On a test site, selected from annex B, the equipment shall be placed at the specified height on a non-conducting support and in a position which is closest to normal use as declared by the manufacturer.

The transmitter antenna connector shall be connected to an artificial antenna, subclause 6.5.

The test antenna shall be orientated for vertical polarization and the length of the test antenna shall be chosen to correspond to the instantaneous frequency of the measuring receiver, or a suitable broadband antenna may be used.

The output of the test antenna shall be connected to a measuring receiver.

For integral antenna equipment testing, a filter shall be inserted between the test antenna and the measuring receiver. For the measurement of spurious emissions below the second harmonic of the carrier frequency the filter used shall be a high Q (notch) filter centred on the transmitter carrier frequency and attenuating this signal by at least 30 dB. For the measurement of spurious emissions at and above the second harmonic of the carrier frequency the filter used shall be a high pass filter with a stop band rejection exceeding 40 dB and the cut off frequency of this high pass filter shall be approximately 1,5 times the transmitter carrier frequency.

The transmitter shall be switched on without modulation, and the measuring receiver shall be tuned over the frequency range 30 MHz to 2 GHz, except for the channel on which the transmitter is intended to operate and its adjacent channels.

At each frequency at which a spurious component is detected:

a) The test antenna shall be raised and lowered through the specified range of heights until a maximum signal level is detected on the measuring receiver.

b) The transmitter shall be rotated through 360° in the horizontal plane, until the maximum signal level is detected by the measuring receiver.

c) The maximum signal level detected by the measuring receiver shall be noted.

d) The transmitter shall be replaced by a substitution antenna as defined in annex E.

e) The substitution antenna shall be orientated for vertical polarization and the length of the substitution antenna shall be adjusted to correspond to the frequency of the spurious component detected.

f) The substitution antenna shall be connected to a calibrated signal generator.

g) The frequency of the calibrated signal generator shall be set to the frequency of the spurious component detected.

h) The input attenuator setting of the measuring receiver shall be adjusted in order to increase the sensitivity of the measuring receiver, if necessary.

i) The test antenna shall be raised and lowered through the specified range of heights to ensure that the maximum signal is received.

j) The input signal to the substitution antenna shall be adjusted to the level that produces a level detected by the measuring receiver that is equal to the level noted while the spurious component was measured, corrected for the change of input attenuator setting of the measuring receiver.

k) The input level to the substitution antenna shall be recorded as power level, corrected for the change of input attenuator setting of the measuring receiver.

l) The measurement shall also be taken with the test antenna and the substitution antenna orientated for horizontal polarization.

m) The effective radiated power of the spurious component is the larger of the two power levels recorded for that spurious component at the input to the substitution antenna, corrected to compensate for the gain of the antenna if necessary.

n) The measurements shall be repeated with the transmitter in stand-by mode.

8.9.3 Limits

With the transmitter in stand-by mode the cabinet radiation and spurious emissions shall not exceed 2 nW.

With the transmitter in operation the cabinet radiation and spurious emissions shall not exceed 0,25 µW.

8.10 Residual modulation of the transmitter

8.10.1 Definition

The residual modulation of the transmitter is the ratio, in dB, of the demodulated RF signal in the absence of wanted modulation, to the demodulated RF signal produced when the normal test modulation is applied.

8.10.2 Method of measurement

The normal test modulation defined in subclause 6.4 shall be applied to the transmitter. The high frequency signal produced by the transmitter shall be applied, via an appropriate coupling device, to a linear demodulator with a de-emphasis network of 6 dB per octave. The time constant of this de-emphasis network shall be at least 750 µsec.

Precautions shall be taken to avoid the effects of emphasizing the low audio frequencies produced by internal noise.

The signal shall be measured at the demodulator output using an rms voltmeter.

The modulation shall then be switched off and the level of the residual audio frequency signal at the output shall be measured again.

8.10.3 Limit

The residual modulation shall not exceed -40 dB.

8.11 Transient frequency behaviour of the transmitter

8.11.1 Definitions

The transient frequency behaviour of the transmitter is the variation in time of the difference between the instantaneous transmitter frequency and the nominal frequency of the transmitter when the RF output power is switched on and off:

- t_{on}: according to the method of measurement described in subclause 8.11.2 the switch-on instant t_{on} of a transmitter is defined by the condition when the output power, measured at the antenna terminal, exceeds 0,1 % of the nominal power;
- t_1: period of time starting at t_{on} and finishing according to table 2;
- t_2: period of time starting at the end of t_1 and finishing according to table 2;
- t_{off}: switch-off instant defined by the condition when the nominal power falls below 0,1 % of the nominal power;
- t_3: period of time finishing at t_{off} and starting according to table 2.

Table 2

t_1 (ms)	5,0
t_2 (ms)	20,0
t_3 (ms)	5,0
NOTE: During the periods t_1 and t_3 the frequency difference shall not exceed the value of 25 kHz. During the period t_2 the frequency difference shall not exceed the value of 12,5 kHz.	

8.11.2 Method of measurement

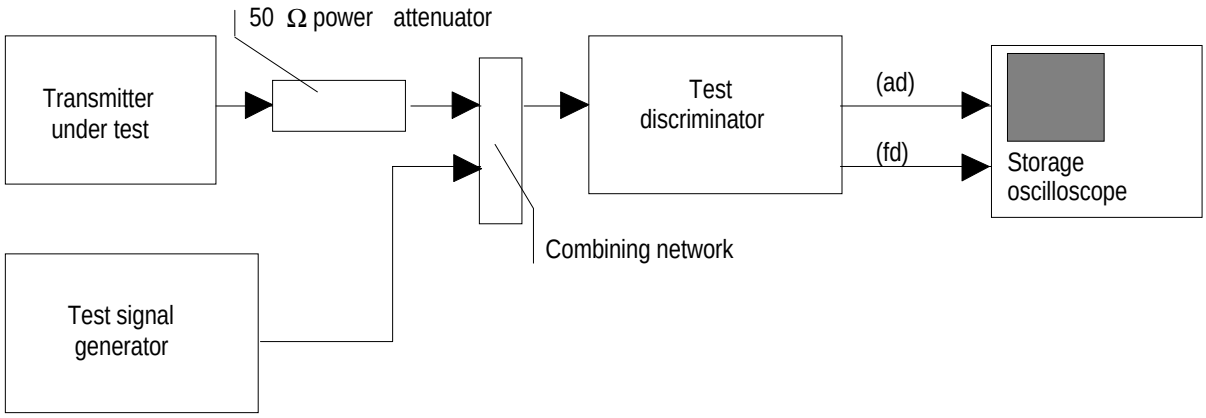


Figure 3: Measurement arrangement

Two signals shall be connected to the test discriminator via a combining network.

The transmitter shall be connected to a 50 Ω power attenuator.

The output of the power attenuator shall be connected to the test discriminator via one input of the combining network.

A test signal generator shall be connected to the second input of the combining network.

The test signal shall be adjusted to the nominal frequency of the transmitter.

The test signal shall be modulated by a frequency of 1 kHz with a deviation of ± 25 kHz.

The test signal level shall be adjusted to correspond to 0,1 % of the power of the transmitter under test measured at the input of the test discriminator. This level shall be maintained throughout the measurement.

The amplitude difference (ad) and the frequency difference (fd) output of the test discriminator shall be connected to a storage oscilloscope.

The storage oscilloscope shall be set to display the channel corresponding to the frequency difference (fd) input up to ± 1 channel frequency difference, corresponding to the relevant channel separation from the nominal frequency.

The storage oscilloscope shall be set to a sweep rate of 10 ms/division and set so that the triggering occurs at 1 division from the left edge of the display.

The display shows the 1 kHz test signal continuously.

The storage oscilloscope shall then be set to trigger on the channel corresponding to the amplitude difference (ad) input at a low input level, rising.

The transmitter shall then be switched on, without modulation, to produce the trigger pulse and a picture on the display.

The result of the change in the ratio of power between the test signal and the transmitter output, due to the capture ratio of the test discriminator, produces two separate sides on the picture, one showing the 1 kHz test signal, the other the frequency difference of the transmitter versus time:

- the moment when the 1 kHz test signal is completely suppressed is considered to provide t_{on};
- the periods of time t_1 and t_2 as defined in the table shall be used to define the appropriate template;
- the result shall be recorded as frequency difference versus time;
- the transmitter shall remain switched on.

The storage oscilloscope shall be set to trigger on the channel corresponding to the amplitude difference (ad) input at a high input level, decaying and set so that the triggering occurs at 1 division from the right edge of the display:

- the transmitter shall then be switched off;
- the moment when the 1 kHz test signal starts to rise is considered to provide t_{off};
- the period of time t_3 as defined in table 2 shall be used to define the appropriate template;
- the result shall be recorded as frequency difference versus time.

8.11.3 Limits

- During the period of time t_1 the frequency difference shall not exceed the value of 25 kHz.
- During the period of time t_2 the frequency difference shall not exceed the value of 12,5 kHz.
- The frequency difference, after t_2, shall not exceed the value of 1,5 kHz.
- Before the start of t_3 the frequency difference shall not exceed the value of 1,5 kHz.
- During the period of time t_3 the frequency difference shall not exceed the value of 25 kHz.

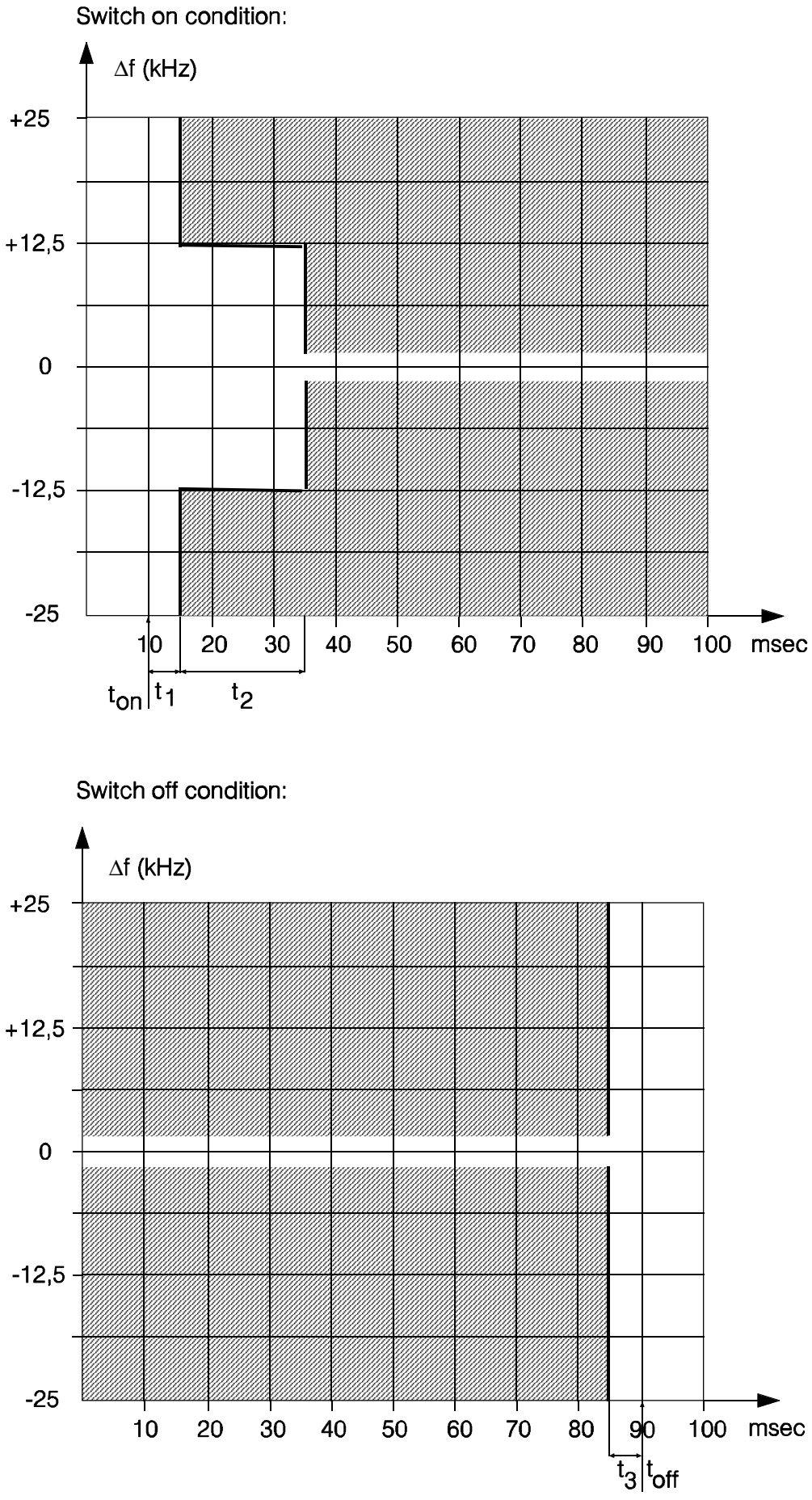

Switch on condition:
Δf (kHz)
+25
+12,5
0
-12,5
-25
10
20
30
40
50
60
70
80
90
100
msec
t_{on}
t_1
t_2
Switch off condition:
Δf (kHz)
+25
+12,5
0
-12,5
-25
10
20
30
40
50
60
70
80
90
100
msec
t_3
t_{off}

Figure 4

9 Receiver

9.1 Harmonic distortion and rated audio frequency output power

9.1.1 Definition

The harmonic distortion at the receiver output is defined as the ratio, expressed as a percentage, of the total rms voltage of all the harmonic components of the modulation audio frequency to the total rms voltage of the signal delivered by the receiver.

The rated audio frequency output power is the value stated by the manufacturer to be the maximum power available at the output, for which all the requirements of the present document are met.

9.1.2 Methods of measurement

Test signals at a level of +100 dBµV, at a carrier frequency equal to the nominal frequency of the receiver and modulated by the normal test modulation (subclause 6.4) shall be applied in succession to the receiver input under the conditions specified in subclause 6.1.

For each measurement, the receiver's audio frequency volume control shall be set so as to obtain, in a resistive load which simulates the receiver's operating load, the rated audio frequency output power (subclause 9.1.1). The value of this load shall be stated by the manufacturer.

Under normal test conditions (subclause 6.10) the test signal shall be modulated successively at 300 Hz, 500 Hz and 1 kHz with a constant modulation index of 3 (ratio between the frequency deviation and the modulation frequency). The harmonic distortion and audio frequency output power shall be measured at all the frequencies specified above.

Under extreme test conditions (subclause 6.11.1 and subclause 6.11.2 applied simultaneously), the tests shall be made at the receiver's nominal frequency and at the nominal frequency ± 1,5 kHz. For these tests, the modulation shall be 1 kHz and the frequency deviation shall be ± 3 kHz.

9.1.3 Limits

The rated audio frequency output power shall be at least:

- 0,2 W in a loudspeaker;
- 1 mW in the handset earphone.

The harmonic distortion shall not exceed 10 %.

9.2 Audio frequency response

9.2.1 Definition

The audio frequency response is the variation in the receiver's audio frequency output level as a function of the modulating frequency of a received radio frequency signal modulated with constant deviation.

9.2.2 Method of measurement

A test signal of +60 dBµV (e.m.f.), at a carrier frequency equal to the nominal frequency of the receiver and modulated with normal test modulation (subclause 6.4), shall be applied to the receiver antenna port under the conditions specified in subclause 6.2.

The receiver's audio frequency power control shall be set so as to produce a power level of at least 50 % of the rated output power (subclause 9.1). This setting shall remain unchanged during the test.

The frequency deviation shall then be reduced to 1 kHz and the audio output is the reference point in figure 5 (1 kHz corresponds to 0 dB).

The frequency deviation shall remain constant while the modulation frequency is varied between 300 Hz and 3 kHz and the output level shall then be measured.

The measurement shall be repeated with a test signal at frequencies 1,5 kHz above and below the nominal frequency of the receiver.

The test shall be carried out on one channel only (see subclause 6.7).

9.2.3 Limits

The audio frequency response shall not deviate by more than +1 dB or -3 dB from a characteristic giving the output level as a function of the audio frequency, decreasing by 6 dB per octave and passing through the measured point at 1 kHz (figure 5).

Certified Intrinsically Safe equipment need not comply with the limits below 700 Hz.

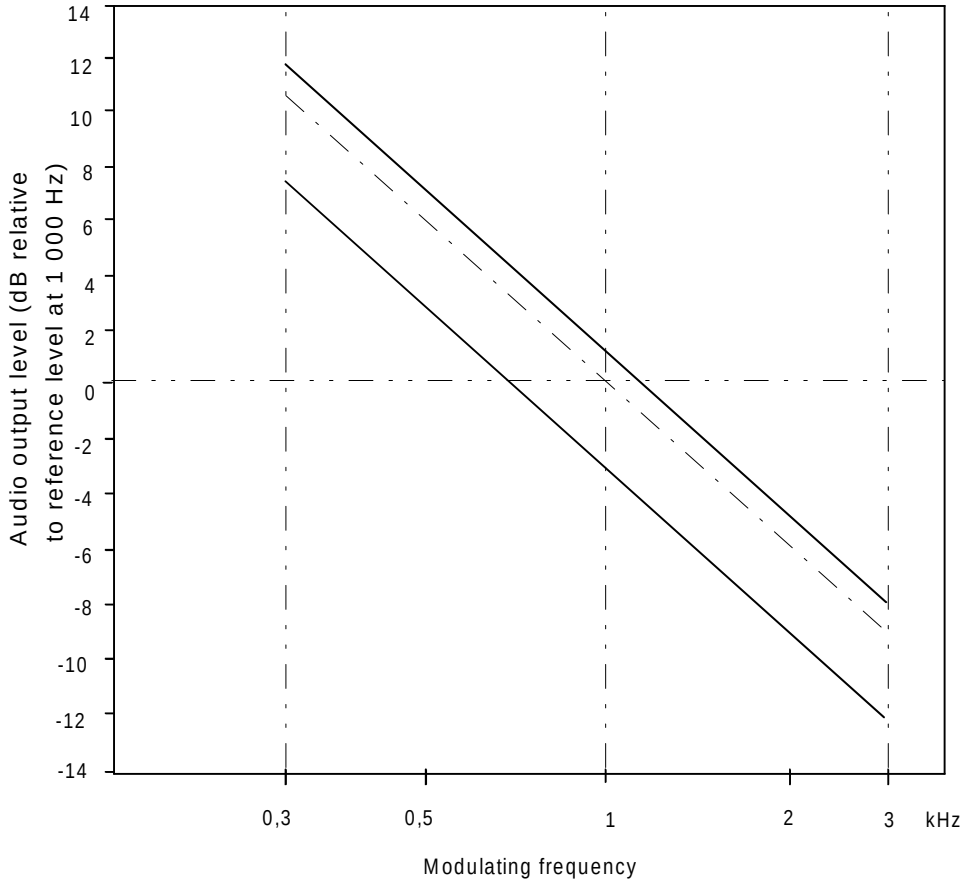


Figure 5: Audio frequency response

9.3 Maximum usable sensitivity

9.3.1 Definition

The maximum usable sensitivity of the receiver is the minimum level of the signal (emf) at the nominal frequency of the receiver which, when applied to the receiver input with normal test modulation (subclause 6.4), will produce:

- in all cases, an audio frequency output power of at least 50 % of the rated output power (subclause 9.1); and
- a SINAD ratio of 20 dB, measured at the receiver output through a psophometric telephone filtering network such as described in ITU-T Recommendation P.53 [6].

9.3.2 Method of measurement

The test shall be performed on the lowest frequency channel, the highest frequency channel and on channel 16.

A test signal at a carrier frequency equal to the nominal frequency of the receiver, modulated by the normal test modulation (subclause 6.4) shall be applied to the receiver input. An audio frequency load and a measuring instrument for measuring the SINAD ratio (through a psophometric network as specified in subclause 9.3.1) shall be connected to the receiver output terminals.

The level of the test signal shall be adjusted until a SINAD ratio of 20 dB is obtained, using the psophometric network and with the receiver's audio frequency power control adjusted to produce at least 50 % of the rated output power. Under these conditions, the level of the test signal at the input is the value of the maximum usable sensitivity.

The measurements shall be taken under normal test conditions (subclause 6.10) and under extreme test conditions (subclause 6.11.1 and subclause 6.11.2 applied simultaneously).

A receiver output power variation of ± 3 dB relative to 50 % of the rated output power may be allowed for sensitivity measurements under extreme test conditions.

9.3.3 Limits

The maximum usable sensitivity shall not exceed +6 dBµV (emf) under normal test conditions and +12 dBµV (emf) under extreme test conditions.

9.4 Co-channel rejection

9.4.1 Definition

The co-channel rejection is a measure of the capability of the receiver to receive a wanted modulated signal without exceeding a given degradation due to the presence of an unwanted modulated signal, both signals being at the nominal frequency of the receiver.

9.4.2 Method of measurement

The two input signals shall be connected to the receiver via a combining network (subclause 6.2). The wanted signal shall have normal test modulation (subclause 6.4). The unwanted signal shall be modulated by 400 Hz with a deviation of ± 3 kHz. Both input signals shall be at the nominal frequency of the receiver under test and the measurement repeated for displacements of the unwanted signal of up to ± 3 kHz.

The wanted input signal level shall be set to the value corresponding to the maximum usable sensitivity as measured in subclause 9.3. The amplitude of the unwanted input signal shall then be adjusted until the SINAD ratio (psophometrically weighted) at the output of the receiver is reduced to 14 dB.

The co-channel rejection ratio shall be expressed as the ratio in dB of the level of the unwanted signal to the level of the wanted signal at the receiver input for which the specified reduction in SINAD ratio occurs.

9.4.3 Limit

The co-channel rejection ratio, at any frequency of the unwanted signal within the specified range, shall be between -10 dB and 0 dB.

9.5 Adjacent channel selectivity

9.5.1 Definition

The adjacent channel selectivity is a measure of the capability of the receiver to receive a wanted modulated signal without exceeding a given degradation due to the presence of an unwanted modulated signal which differs in frequency from the wanted signal by 25 kHz.

9.5.2 Method of measurement

The test shall be performed on the lowest frequency channel, the highest frequency channel and on channel 16.

The two input signals shall be applied to the receiver input via a combining network (subclause 6.2). The wanted signal shall be at the nominal frequency of the receiver and shall have normal test modulation (subclause 6.4). The unwanted signal shall be modulated by 400 Hz with a deviation of ± 3 kHz, and shall be at a frequency 25 kHz above that of the wanted signal.

The wanted input signal level shall be set to the value corresponding to the maximum usable sensitivity as measured in subclause 9.3. The amplitude of the unwanted input signal shall then be adjusted until the SINAD ratio at the receiver output, psophometrically weighted, is reduced to 14 dB. The measurement shall be repeated with an unwanted signal at a frequency 25 kHz below that of the wanted signal.

The adjacent channel selectivity shall be expressed as the lower value of the ratios in dB for the upper and lower adjacent channels of the level of the unwanted signal to the level of the wanted signal.

The test shall then be repeated under extreme test conditions (subclause 6.11.1 and subclause 6.11.2 applied simultaneously) with the wanted signal set to the value corresponding to the maximum usable sensitivity under these conditions.

9.5.3 Limits

The adjacent channel selectivity shall be not less than 70 dB under normal test conditions and not less than 60 dB under extreme test conditions.

9.6 Spurious response rejection

9.6.1 Definition

The spurious response rejection is a measure of the capability of the receiver to discriminate between the wanted modulated signal at the nominal frequency and an unwanted signal at any other frequency at which a response is obtained.

9.6.2 Method of measurement

Two input signals shall be applied to the receiver input via a combining network (subclause 6.2). The wanted signal shall be at the nominal frequency of the receiver and shall have normal test modulation (subclause 6.4).

The unwanted signal shall be modulated by 400 Hz with a deviation of ± 3 kHz.

The wanted input signal level shall be set to the value corresponding to the maximum usable sensitivity as measured in subclause 9.3. The amplitude of the unwanted input signal shall be adjusted to an emf of +86 dBµV. The frequency shall then be swept over the frequency range from 100 kHz to 2 000 MHz.

At any frequency at which a response is obtained, the input level shall be adjusted until the SINAD ratio psophometrically weighted, is reduced to 14 dB.

The spurious response rejection ratio shall be expressed as the ratio in dB between the unwanted signal and the wanted signal at the receiver input when the specified reduction in the SINAD ratio is obtained.

9.6.3 Limit

At any frequency separated from the nominal frequency of the receiver by more than 25 kHz, the spurious response rejection ratio shall be not less than 70 dB.

9.7 Intermodulation response

9.7.1 Definition

The intermodulation response is a measure of the capability of a receiver to receive a wanted modulated signal without exceeding a given degradation due to the presence of two or more unwanted signals with a specific frequency relationship to the wanted signal frequency.

9.7.2 Method of measurement

Three signal generators, A, B and C shall be connected to the receiver via a combining network (subclause 6.2). The wanted signal, represented by signal generator A shall be at the nominal frequency of the receiver and shall have normal test modulation (subclause 6.4). The unwanted signal from signal generator B shall be unmodulated and adjusted to the frequency 50 kHz above (or below) the nominal frequency of the receiver. The second unwanted signal from signal generator C shall be modulated by 400 Hz with a deviation of ± 3 kHz, and adjusted to a frequency 100 kHz above (or below) the nominal frequency of the receiver.

The wanted input signal shall be set to a value corresponding to the maximum usable sensitivity as measured in subclause 9.3. The amplitude of the two unwanted signals shall be maintained equal and shall be adjusted until the SINAD ratio at the receiver output, psophometrically weighted, is reduced to 14 dB. The frequency of signal generator B shall be adjusted slightly to produce the maximum degradation of the SINAD ratio. The level of the two unwanted test signals shall be readjusted to restore the SINAD ratio of 14 dB. The intermodulation response ratio shall be expressed as the ratio in dB between the two unwanted signals and the wanted signal at the receiver input, when the specified reduction in the SINAD ratio is obtained.

9.7.3 Limit

The intermodulation response ratio shall not be less than 68 dB.

9.8 Blocking or desensitization

9.8.1 Definition

Blocking is a change (generally a reduction) in the wanted output power of the receiver or a reduction of the SINAD ratio due to an unwanted signal on another frequency.

9.8.2 Method of measurement

Two input signals shall be applied to the receiver via a combining network (subclause 6.2). The modulated wanted signal shall be at the nominal frequency of the receiver and shall have normal test modulation (subclause 6.4). Initially the unwanted signal shall be switched off and the wanted signal set to the value corresponding to the maximum usable sensitivity.

The output power of the wanted signal shall be adjusted, where possible, to 50 % of the rated output power and in the case of stepped volume controls, to the first step that provides an output power of at least 50 % of the rated output power. The unwanted signal shall be unmodulated and the frequency shall be swept between +1 MHz and +10 MHz, and also between -1 MHz and -10 MHz, relative to the nominal frequency of the receiver. The input level of the unwanted signal, at all frequencies in the specified ranges, shall be so adjusted that the unwanted signal causes:

a) a reduction of 3 dB in the output level of the wanted signal; or

b) a reduction to 14 dB of the SINAD ratio at the receiver output using a psophometric telephone filtering network such as described in ITU-T Recommendation P.53 [6] whichever occurs first. This level shall be noted.

9.8.3 Limit

The blocking level for any frequency within the specified ranges, shall be not less than 90 dBµV (e.m.f), except at frequencies on which spurious responses are found (subclause 9.6).

9.9 Conducted spurious emissions

9.9.1 Definition

Conducted spurious emissions from the receiver are components at any frequency, present at the receiver input port.

9.9.2 Method of measurement

The level of spurious emissions shall be measured as the power level at the antenna.

Conducted spurious radiations shall be measured as the power level of any discrete signal at the input terminals of the receiver. The receiver input terminals are connected to a spectrum analyser or selective voltmeter having an input impedance of 50 Ω and the receiver is switched on.

If the detecting device is not calibrated in terms of power input, the level of any detected components shall be determined by a substitution method using a signal generator.

The measurements shall extend over the frequency range of 9 kHz to 2 GHz.

9.9.3 Limit

The power of any spurious radiation shall not exceed 2 nW at any frequency in the range between 9 kHz and 2 Ghz.

9.10 Radiated spurious emissions

9.10.1 Definition

Radiated spurious emissions from the receiver are components at any frequency radiated by the equipment cabinet and the structure.

Integral antenna equipment shall be tested with the normal antenna fitted.

9.10.2 Method of measurements

On a test site, selected from annex B, the equipment shall be placed at the specified height on a non-conducting support and in a position which is closest to normal use as declared by the manufacturer.

The test antenna shall be orientated for vertical polarization and the length of the test antenna shall be chosen to correspond to the instantaneous frequency of the measuring receiver, or a suitable broadband antenna may be used.

The output of the test antenna shall be connected to a measuring receiver.

The receiver shall be switched on without modulation, and measuring receiver shall be tuned over the frequency range 30 MHz to 2 GHz.

At each frequency at which a spurious component is detected:

a) The test antenna shall be raised and lowered through the specified range of heights until a maximum signal level is detected on the measuring receiver.

b) The receiver shall be rotated through 360° in the horizontal plane, until the maximum signal level is detected by the measuring receiver.

c) The maximum signal level detected by the measuring receiver shall be noted.

d) The receiver shall be replaced by a substitution antenna as defined in annex B.

e) The substitution antenna shall be orientated for vertical polarization and the length of the substitution antenna shall be adjusted to correspond to the frequency of the spurious component detected.

f) The substitution antenna shall be connected to a calibrated signal generator.

g) The frequency of the calibrated signal generator shall be set to the frequency of the spurious component detected.

h) The input attenuator setting of the measuring receiver shall be adjusted in order to increase the sensitivity of the measuring receiver, if necessary.

i) The test antenna shall be raised and lowered through the specified range of heights to ensure that the maximum signal is received.

j) The input signal to the substitution antenna shall be adjusted to the level that produces a level detected by the measuring receiver that is equal to the level noted while the spurious component was measured, corrected for the change of input attenuator setting of the measuring receiver.

k) The input level to the substitution antenna shall be recorded as power level, corrected for the change of input attenuator setting of the measuring receiver.

l) The measurement shall also be taken with the test antenna and the substitution antenna orientated for horizontal polarization.

m) The effective radiated power of the spurious component is the larger of the two power levels recorded for that spurious component at the input to the substitution antenna, corrected to compensate for the gain of the antenna if necessary.

9.10.3 Limit

The power of any spurious radiation shall not exceed 2 nW at any frequency in the range between 30 MHz and 2 GHz.

9.11 Receiver noise and hum level

9.11.1 Definition

The receiver noise and hum level is defined as the ratio, in dB, of the audio frequency power of the noise and hum resulting from spurious effects of the power supply system or from other causes, to the audio frequency power produced by a high frequency signal of average level, modulated by the normal test modulation and applied to the receiver input.

9.11.2 Method of measurement

A test signal with a level of +30 dBµV (e.m.f.) at a carrier frequency equal to the nominal frequency of the receiver, and modulated by the normal test modulation specified in subclause 6.4, shall be applied to the receiver input. An audio frequency load shall be connected to the output terminals of the receiver. The audio frequency power control shall be set so as to produce the rated output power level conforming to subclause 9.1.

The output signal shall be measured by an rms voltmeter having a -6 dB bandwidth of at least 20 kHz. The modulation shall then be switched off and the audio frequency output level measured again.

9.11.3 Limit

The receiver noise and hum level shall not exceed -40 dB, relative to the modulated signal.

9.12 Squelch operation

9.12.1 Definition

The purpose of the squelch facility is to mute the receiver audio output signal when the level of the signal at the receiver input is less than a given value.

9.12.2 Method of measurement

The following procedure shall be followed:

a) With the squelch facility switched off, a test signal of +30 dBµV, at a carrier frequency equal to the nominal frequency of the receiver and modulated by the normal test modulation specified in subclause 6.4, shall be applied to the input terminals of the receiver. An audio frequency load and a psophometric filtering network (subclause 9.3.1) shall be connected to the output terminals of the receiver. The receiver's audio frequency power control shall be set so as to produce the rated output power defined in subclause 9.1.

   The output signal shall be measured with the aid of an rms voltmeter.

   The input signal shall then be suppressed, the squelch facility switched on and the audio frequency output level measured again.

b) With the squelch facility switched off again, a test signal modulated by the normal test modulation shall be applied to the receiver input at a level of +6 dBµV (e.m.f.) and the receiver shall be set to produce at least 50 % of the rated output power. The level of the input signal shall then be reduced and the squelch facility shall be switched on. The input signal shall then be increased until the above-mentioned output power is reached. The SINAD ratio and the input level shall then be measured.

c) (Applicable only to equipment with continuously adjustable squelch control) with the squelch facility switched off, a test signal with normal test modulation shall be applied to the receiver input at a level of +6 dBµV (e.m.f.), and the receiver shall be set to produce at least 50 % of the rated audio output power. The level of the input signal shall then be reduced and the squelch facility shall be switched on. The squelch shall then be at its maximum position and the level of the input signal increased until the output power returns to at least 50 % of the rated audio output power.

9.12.3 Limits

Under the conditions specified in a) subclause 9.11.2, the audio frequency output power shall not exceed -40 dB relative to the rated output power.

Under the conditions specified in b) subclause 9.11.2, the input level shall not exceed +6 dBµV (e.m.f.).

Under the conditions specified in c) subclause 9.11.2, the input signal shall not exceed +6 dBµV (e.m.f.) when the control is set at maximum.

9.13 Squelch hysteresis

9.13.1 Definition

Squelch hysteresis is the difference in dB between the receiver input signal levels at which the squelch opens and closes.

9.13.2 Method of measurement

If there is any squelch control on the exterior of the equipment it shall be placed in its maximum muted position. With the squelch facility switched on, an unmodulated input signal at a carrier frequency equal to the nominal frequency of the receiver shall be applied to the input of the receiver at a level sufficiently low to avoid opening the squelch. The input signal shall be increased to the level just opening the squelch. This input level shall be recorded. With the squelch still open, the level of the input signal shall be slowly decreased until the squelch mutes the receiver audio output again, the hysteresis being the difference between this level and the previously recorded level.

9.13.3 Limit

The squelch hysteresis shall be between 3 dB and 6 dB.

Annex A (normative): Measuring receiver for adjacent channel power measurement

A.1 Power measuring receiver specification

The power measuring receiver consists of a mixer, an Intermediate Frequency (IF) filter, an oscillator, an amplifier, a variable attenuator and an rms value indicator. Instead of the variable attenuator with the rms value indicator it is also possible to use an rms voltmeter calibrated in dB. The technical characteristics of the power measuring receiver are given below (see also ITU-R Recommendation SM 332-4 [7]).

A.1.1 IF filter

The IF filter shall be within the limits of the following selectivity characteristics.

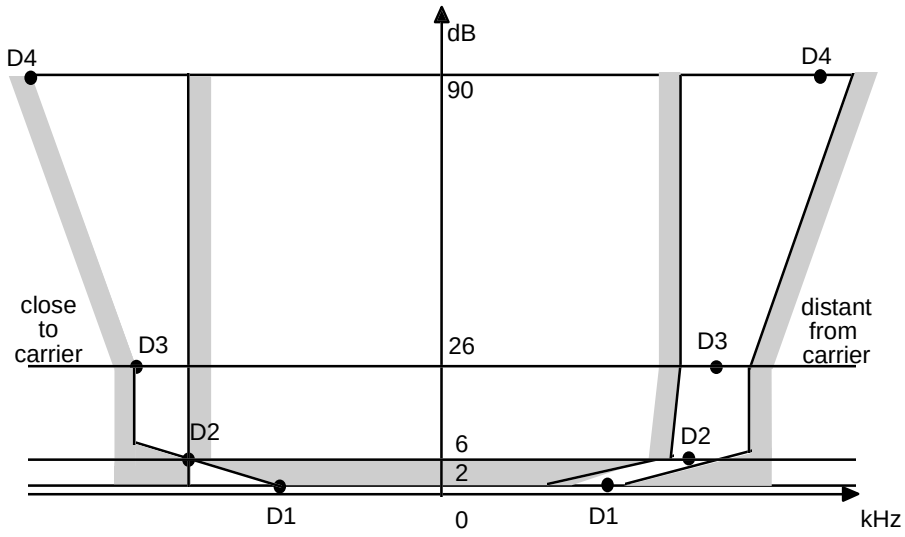


Figure A.1: IF filter characteristics

The selectivity characteristics shall keep the frequency separations shown in table A.1 from the nominal centre frequency of the adjacent channel.

Table A.1: Selectivity characteristic

Frequency separation of filter curve from nominal centre frequency of adjacent channel (kHz)			
D1	D2	D3	D4
5,0	8,0	9,25	13,25

The attenuation points shall not exceed the tolerances shown in tables A.2 and A.3.

Table A.2: Attenuation points close to carrier

Tolerance range (kHz)			
D1	D2	D3	D4
+3,1	± 0,1	-1,35	-5,35

Table A.3: Attenuation points distant from the carrier

Tolerance range (kHz)			
D1	D2	D3	D4
± 3,5	± 3,5	± 3,5	+3,5 -7,5

The minimum attenuation of the filter outside the 90 dB attenuation points shall be equal to or greater than 90 dB.

A.1.2 Attenuation indicator

The attenuation indicator shall have a minimum range of 80 dB and a reading accuracy of 1 dB. With a view to future regulations an attenuation of 90 dB or more is recommended.

A.1.3 Rms value indicator

The instrument shall accurately indicate non-sinusoidal signals in ratio of up to 10:1 between peak value and rms value.

A.1.4 Oscillator and amplifier

The oscillator and the amplifier shall be designed in such a way that the measurement of the adjacent channel power of a low-noise unmodulated transmitter, whose self-noise has a negligible influence on the measurement result, yields a measured value of less than -90 dB.

Annex B (normative): Radiated measurements

B.1 Test sites and general arrangements for measurements involving the use of radiated fields

B.1.1 Outdoor test site

The outdoor test site shall be on a reasonably level surface or ground. At one point on the site, a ground plane of at least 5 m diameter shall be provided. In the middle of this ground plane, a non-conducting support, capable of rotation through 360° in the horizontal plane, shall be used to support the test sample at 1,5 m above the ground plane. The test site shall be large enough to allow the erection of a measuring or transmitting antenna at a distance of half a wavelength or 3 m whichever is the greater. The distance actually used shall be recorded with the results of the tests carried out on the site.

Sufficient precautions shall be taken to ensure that reflections from extraneous objects adjacent to the site and ground reflections do not degrade the measurements results.

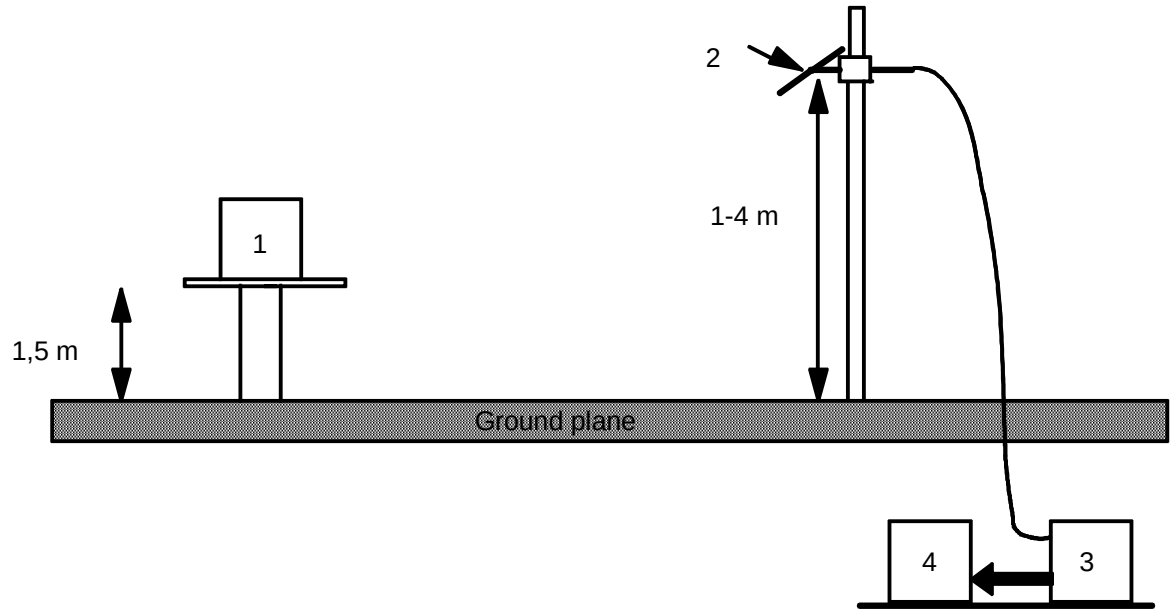


Figure B.1

Key:
1 EUT
2 Test antenna
3 High pass filter (necessary for strong fundamental Tx radiation)
4 Spectrum analyser or measuring receiver

B.1.2 Test antenna

The test antenna is used to detect the radiation from both the test sample and the substitution antenna, when the site is used for radiation measurements; where necessary, it is used as a transmitting antenna, when the site is used for the measurement of receiver characteristics.

This antenna is mounted on a support such as to allow the antenna to be used in either horizontal or vertical polarization and for the height of its centre above ground to be varied over the range 1 to 4 m. Preferably a test antenna with pronounced directivity should be used. The size of the test antenna along the measurement axis shall not exceed 20 % of the measuring distance.

For receiver and transmitter radiation measurements, the test antenna is connected to a measuring receiver, capable of being tuned to any frequency under investigation and of measuring accurately the relative levels of signals at its input. For receiver radiated sensitivity measurements the test antenna is connected to a signal generator.

B.1.3 Substitution antenna

When measuring in the frequency range up to 1 GHz the substitution antenna shall be a half wavelength dipole, resonant at the frequency under consideration, or a shortened dipole, calibrated to the half wavelength dipole. When measuring in the frequency range above 4 GHz a horn radiator shall be used. For measurements between 1 and 4 GHz either a half wavelength dipole or a horn radiator may be used. The centre of this antenna shall coincide with the reference point of the test sample it has replaced. This reference point shall be the volume centre of the sample when its antenna is mounted inside the cabinet, or the point where an external antenna is connected to the cabinet.

The distance between the lower extremity of the dipole and the ground shall be at least 0,3 m.

The substitution antenna shall be connected to a calibrated signal generator when the site is used for spurious radiation measurements and transmitter effective radiated power measurements. The substitution antenna shall be connected to a calibrated measuring receiver when the site is used for the measurement of receiver sensitivity.

The signal generator and the receiver shall be operating at the frequencies under investigation and shall be connected to the antenna through suitable matching and balancing networks.

NOTE: The gain of a horn antenna is generally expressed relative to an isotropic radiator.

B.1.4 Optional additional indoor site

When the frequency of the signals being measured is greater than 80 MHz, use may be made of an indoor site. If this alternative site is used, this shall be recorded in the test report.

The measurement site may be a laboratory room with a minimum area of 6 m by 7 m and at least 2,7 m in height.

Apart from the measuring apparatus and the operator, the room shall be as free as possible from reflecting objects other than the walls, floor and ceiling.

The potential reflections from the wall behind the EUT are reduced by placing a barrier of absorbent material in front of it. The corner reflector around the test antenna is used to reduce the effect of reflections from the opposite wall and from the floor and ceiling in the case of horizontally polarized measurements. Similarly, the corner reflector reduces the effects of reflections from the side walls for vertically polarized measurements. For the lower part of the frequency range (below approximately 175 MHz) no corner reflector or absorbent barrier is needed. For practical reasons, the half-wavelength antenna in figure B.2 may be replaced by an antenna of constant length, provided that this length is between a quarter wavelength and a full wavelength at the frequency of measurement and the sensitivity of the measuring system is sufficient. In the same way the distance of half-wavelength to the apex may be varied.

The test antenna, measuring receiver, substitution antenna and calibrated signal generator are used in a way similar to that of the general method.

To ensure that errors are not caused by the propagation path approaching the point at which phase cancellation between direct and the remaining reflected signals occurs, the substitution antenna shall be moved through a distance of ± 0,1 m in the direction of the test antenna as well as in the two directions perpendicular to this first direction.

If these changes of distance cause a signal change of greater than 2 dB, the test sample should be re-sited until a change of less than 2 dB is obtained.

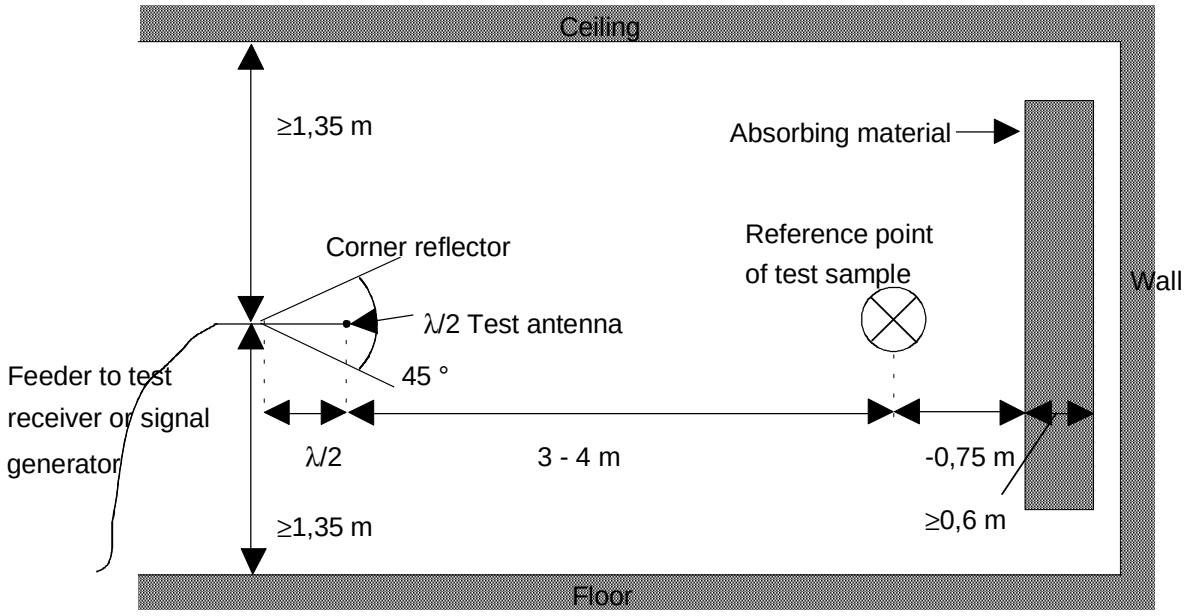


Figure B.2: Indoor site arrangement (shown for horizontal polarization)

B.2 Guidance on the use of radiation test sites

For measurements involving the use of radiated fields, use may be made of a test site in conformity with the requirements of subclause B.1 of this annex. When using such a test site, the following conditions should be observed to ensure consistency of measuring results.

B.2.1 Measuring distance

Evidence indicates that the measuring distance is not critical and does not significantly affect the measuring results, provided that the distance is not less than half wavelength at the frequency of measurement, and the precautions described in this annex are observed. Measuring distances of 3, 5, 10 and 30 m are in common use in European test laboratories.

B.2.2 Test antenna

Different types of test antenna may be used, since performing substitution measurements reduces the effect of the errors on the measuring results.

Height variation of the test antenna over a range of 1 to 4 m is essential in order to find the point at which the radiation is a maximum.

Height variation of the test antenna may not be necessary at the lower frequencies below about 100 MHz.

B.2.3 Substitution antenna

Variations in the measuring results may occur with the use of different types of substitution antenna at the lower frequencies below about 80 MHz.

Where a shortened dipole antenna is used at these frequencies, details of the type of antenna used should be included with the results of the tests carried out on the site. Correction factors shall be taken into account when shortened dipole antennas are used.

B.2.4 Artificial antenna

The dimensions of the artificial antenna used during radiated measurements should be small in relation to the sample under test.

Where possible, a direct connection should be used between the artificial antenna and the test sample.

In cases where it is necessary to use a connecting cable, precautions should be taken to reduce the radiation from this cable by, for example, the use of ferrite cores or double screened cables.

B.2.5 Auxiliary cables

The position of auxiliary cables (power supply and microphone cables etc.) which are not adequately decoupled may cause variations in the measuring results. In order to get reproducible results, cables and wires of auxiliaries should be arranged vertically downwards (through a hole in the non conducting support).

B.2.6 Acoustic measuring arrangement

When carrying out measurements of the maximum usable sensitivity (radiated) of the receiver, the audio output shall be monitored by acoustically coupling the audio signal from the receiver loudspeaker/transducer to the test microphone. On the radiation test site all conducting materials shall be placed below the ground surface and the acoustic signal is conveyed from the receiver to the test microphone in a non-conducting acoustic pipe.

The acoustic pipe shall have an appropriate length. The acoustic pipe shall have an inner diameter of 6 mm and a wall thickness of 1,5 mm. A plastic funnel of a diameter corresponding to the receiver loudspeaker/transducer shall be attached to the receiver surface centred in front of the receiver loudspeaker/transducer. The plastic funnel shall be very soft at the attachment point to the receiver in order to avoid mechanical resonance. The narrow end of the plastic funnel shall be connected to the one end of the acoustic pipe and the test microphone to the other.

B.3 Further optional alternative indoor test site using an anechoic chamber

For radiation measurements when the frequency of the signals being measured is greater than 30 MHz, use may be made of an indoor site being a well-shielded anechoic chamber simulating free space environment. If such a chamber is used, this shall be recorded in the test report.

The test antenna, measuring receiver, substitution antenna and calibrated signal generator are used in a way similar to that of the general method, subclause B.1. In the range between 30 MHz and 100 MHz some additional calibration may be necessary.

An example of a typical measurement site may be an electrically shielded anechoic chamber being 10 m long, 5 m broad and 5 m high.

Walls and ceiling should be coated with RF absorbers of 1 m height.

The base should be covered with absorbing material 1 m thick, and a wooden floor, able to carry test equipment and operators.

A measuring distance of 3 to 5 m in the long middle axis of the chamber can be used for measurements up to 12,75 GHz.

The construction of the anechoic chamber is described in the following subclauses.

B.3.1 Example of the construction of a shielded anechoic chamber

Free-field measurements can be simulated in a shielded measuring chamber where the walls are coated with RF absorbers.

Figure B.3 shows the requirements for shielding loss and wall return loss of such a room.

As dimensions and characteristics of usual absorber materials are critical below 100 MHz (height of absorbers < 1 m, reflection attenuation < 20 dB) such a room is preferably suitable for measurements above 100 MHz.

Figure B.4 shows the construction of a shielded measuring chamber having a base area of 5 m by 10 m and a height of 5 m.

Ceilings and walls are coated with pyramidal formed absorbers approximately 1 m high. The base is covered with absorbers which are able to carry and which forms a sort of floor.

The available internal dimensions of the room are 3 m x 8 m x 3 m, so that a measuring distance of maximum 5 m length in the middle axis of this room is available.

At 100 MHz the measuring distance can be extended up to a maximum of 2 wavelengths.

The floor absorbers reject floor reflections so that the antenna height need not be changed and floor reflection influences need not be considered.

All measuring results can therefore be checked with simple calculations and the measuring tolerances have the smallest possible values due to the simple measuring configuration.

For special measurements it can be necessary to re-introduce floor reflections. Taking away the floor absorbers would mean a removal of approximately 24 m^3 absorber material. Therefore the floor absorbers are covered with metal plates of metallic nets instead.

B.3.2 Influence of parasitic reflections in anechoic chambers

For free-space propagation in the far field condition the correlation $E = Eo(Ro/R)$ is valid for the dependence of the field strength E on the distance R, whereby Eo is the reference field strength in the reference distance Ro.

It is useful to use just this correlation for comparison measurements, as all constants are eliminated with the ratio and neither cable attenuation nor antenna mismatch or antenna dimensions are of importance.

Deviations from the ideal curve can be seen easily if the logarithm of the above equation is used, because the ideal correlation of field strength and distance can then be shown as a straight line and the deviations occurring in practice are clearly visible. This indirect method shows the disturbances due to reflections more readily and is far less problematical than the direct measurement of reflection attenuation.

With an anechoic chamber of the dimensions suggested in subclause B.3 at low frequencies up to 100 MHz there are no far field conditions, and therefore reflections are stronger so that careful calibration is necessary.

In the medium frequency range from 100 MHz to 1 GHz the dependence of the field strength on the distance meets the expectations very well.

In the frequency range of 1 to 12,75 GHz, because more reflections will occur, the dependence of the field strength on the distance will not correlate so closely.

B.3.3 Calibration of the shielded anechoic chamber

Careful calibration of the chamber shall be performed over the range 30 MHz to 12,75 GHz.

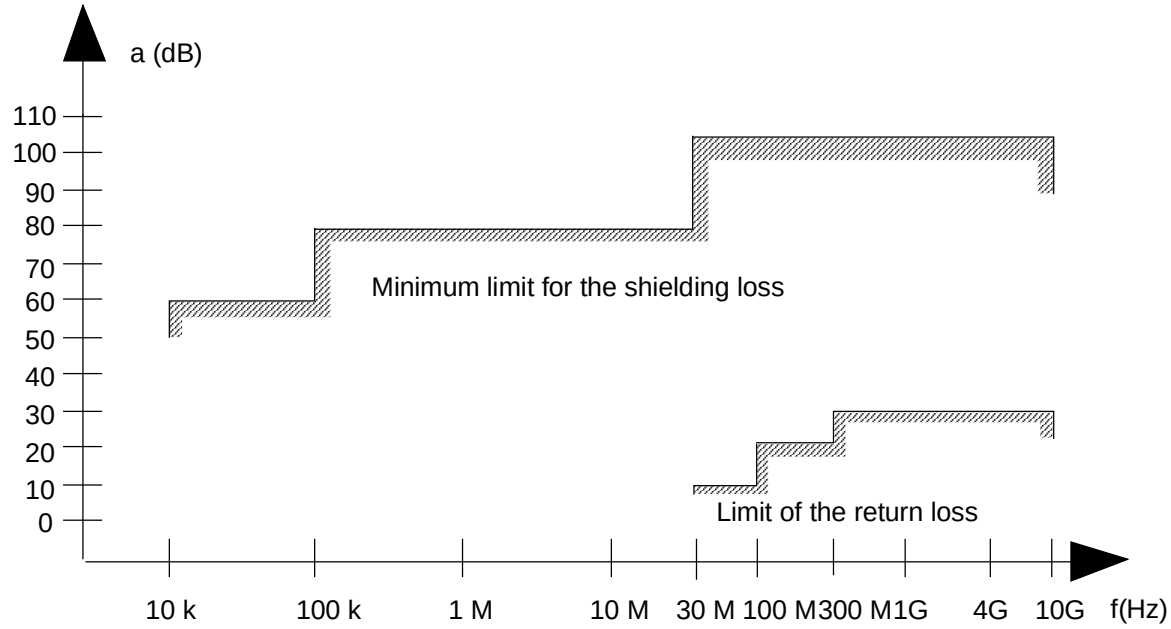
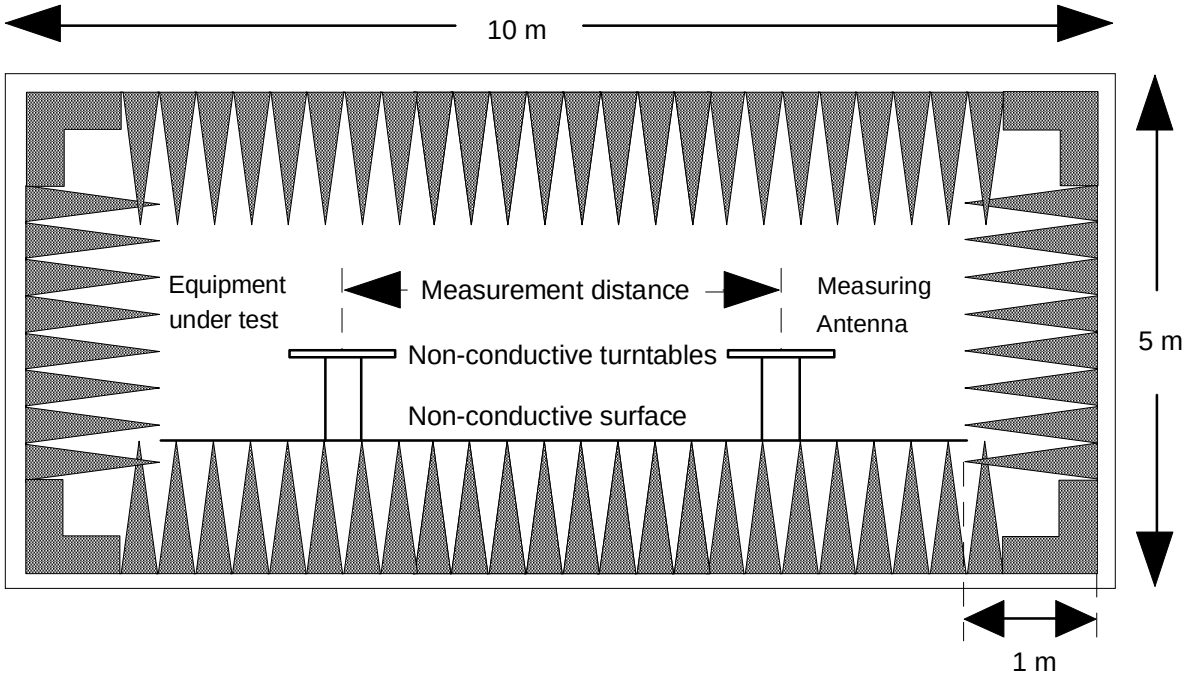


Figure B.3: Specifications for shielding and reflections



Ground plan

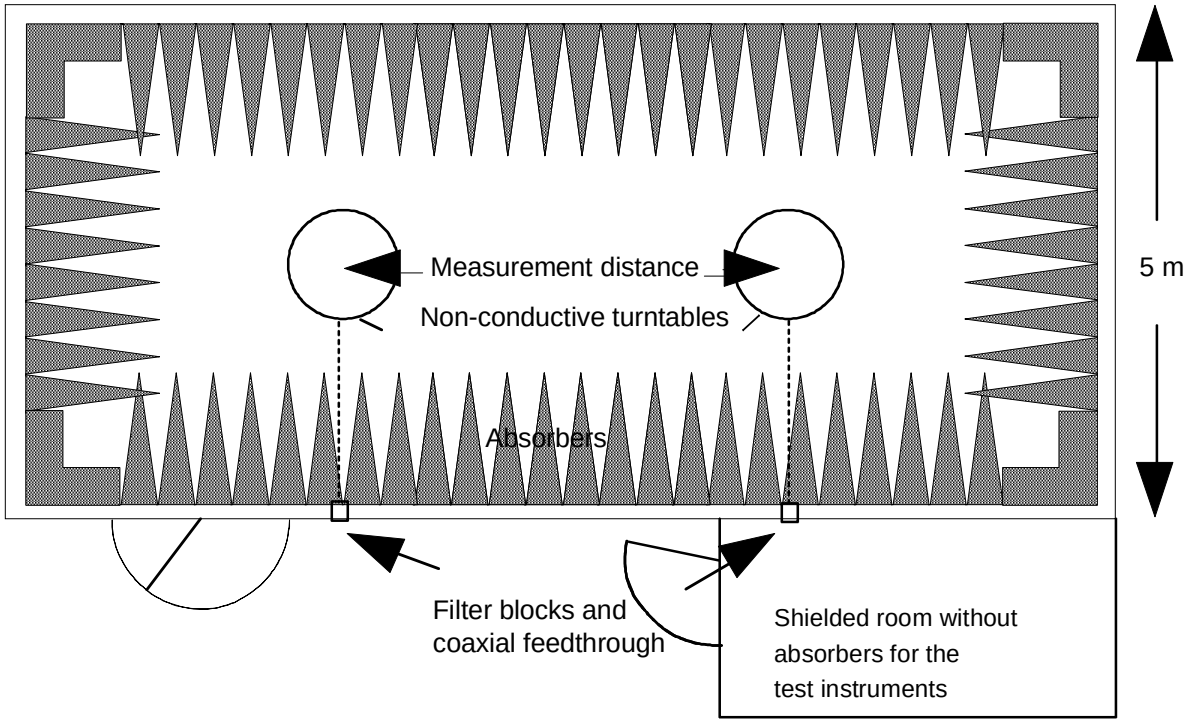


Figure B.4: Example of construction of an anechoic shielded chamber

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